

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	30% of CIA	Total Marks:	100
CO2 Statement:	<i>Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.</i>		
Student Name:	K. Ragavi	Reg. No.:	192524364
Section / Batch:	B. Tech AI&DS/ 2 nd year	Date:	17/07/2026

Code of Conduct

I K. Ragavi (192524364) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K. Ragavi

Instructions

- This test contains 80 Analytical problem-solving questions. Total: 100 Marks.
- When a student answer using an Analytical problem-solving, in evaluation the answer should be with following five requirements: Understanding of problem, select appropriate data structure, Design the solution, analyze, Clarity of representation.
- Each student should write 2 questions. No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

Section / Topic	Focus	Q Nos	Marks
Linked Data Structure, Search Data Structure	List Operations, Binary Search	1	50
Stack Data Structure, Queue Data Structures	Stack Notations, Priority Queue	2	50
GRAND TOTAL:		100 Marks	

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Q6

Linked List

5 marks

Q7

Stack

5 marks

Q8

Linked List Data Structures

5 marks

Q9

Linked List

5 marks

Q10

Queue

5 marks

10/10 answered

✳️ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 int main()
3 {
4     int a[]={10,20,30,40,50};
5     int n = 5, key = 30;
6     int low = 0, high = n-1, mid;
7
8     while(low<=high)
9     {
10         mid=(low + high)/2;
11         if(a[mid]==key)
12         {
13             printf("found at index %d",mid);
14             return 0;
15         }
16         else if(a[mid]<key)
17             low = mid +1;
18         else
19             high=mid-1;
20     }
21     printf("not found");
22     return 0;
23 }
```

Test Input

stdin input (optional)

Output

found at index 2

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Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Stack

5 marks

Q6

Linked List

5 marks

Q7

Stack

5 marks

Q8

Linked List Data Structures

5 marks

Q9

Linked List

5 marks

Q10

Queue

5 marks

return stack[top--];

✳️ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int a[10],n,k,l=0,h,m;
6     scanf("%d",&n);
7     for(int i=0;i<n;i++) scanf("%d",&a[i]);
8     scanf("%d",&k);
9     h=n-1;
10
11     while(l<=h)
12     {
13         m=(l+h)/2;
14         if(a[m]==k) { printf("Found"); return 0; }
15         if(a[m]<k) l=m+1;
16         else h=m-1;
17     }
18     printf("Not Found");
19     return 0;
20 }
```

Test Input

5
10 20 30 40 50
30

Output

Found

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QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q3 Stack 5 marks

Check the snippet and identify the logical error.

```
int stack[5];  
int top;  
void init() {  
    top = 0;  
}
```

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include<stdio.h>  
2 int stack[5];  
3 int top=-1;  
4 int main()  
5 {  
6     printf("stack initialized");  
7     return 0;  
8 }
```

Test Input
stdin input (optional)

Output

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QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List Data Structures
5 marks

Identify the errors in the following snippet.

```
if(ch == ')') {  
    while(stack[top] != ')') {  
        postfix[j++] = pop0();  
    }  
    pop0();  
}
```

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 int main()  
4 {  
5     char stack[10] = {'(', '+', '*', ')'};  
6     int top = 2;  
7  
8     while(stack[top] != '(')  
9     {  
10        printf("%c", stack[top--]);  
11    }  
12    top--;  
13  
14    return 0;  
15 }
```

Test Input
stdin input (optional)

Output

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Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List Data Structures
5 marks

Q9
Linked List
5 marks

```
pop();
} else {
    return 0;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 int main()
4 {
5     char stack[10] = {'['};
6     char ch;
7     int top = 0;
8
9     scanf("%c", &ch);
10
11     if((ch == '[' && stack[top] == '[') ||
12        (ch == ')' && stack[top] == '('))
13         printf("valid");
14     else
15         printf("Invalid");
16
17     return 0;
18 }
```

Test Input
]

Output
Pending manual review

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QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q6 Linked List
5 marks

What is the issue?
temp = head
while temp.next != NULL:
 print(temp.data)
 temp = temp.next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int a[] = {10,20,30}, i = 0;
6
7     while(i < 3)
8         printf("%d ", a[i++]);
9
10    return 0;
11 }
```

Test Input
3
10 20 30

Output
10 20 30

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Q1

Binary Search

5 marks

✓

Q2

Stack

5 marks

✓

Q3

Stack

5 marks

✓

Q4

Stack

5 marks

✓

Q5

Stack

5 marks

✓

Q6

Linked List

5 marks

✓

Q7

Stack

5 marks

✓

Q8

Linked List Data Structures

5 marks

✓

what is the error in the below code?

new.next = head
head = new

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node *next;
7 };
8
9 int main() {
10     struct Node *head = NULL;
11     struct Node *new = malloc(sizeof(struct Node));
12
13     new->data = 10;
14     new->next = head;
15     head = new;
16
17     printf("%d", head->data);
18
19     return 0;
20 }
```

Test Input

stdin input (optional)

Output

10

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Q1

Binary Search

5 marks

✓

Q2

Stack

5 marks

✓

Q3

Stack

5 marks

✓

Q4

Stack

5 marks

✓

Q5

Stack

5 marks

✓

Q6

Linked List

5 marks

✓

Q7

Stack

5 marks

✓

Q8

Linked List Data Structures

5 marks

✓

What is missing in the below pseudocode?

DEQUEUE(Q):
if front == -1:
 print("Underflow")
else:
 return Q[front++]

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include<stdio.h>
2 int main() {
3     int q[5]={10,20,30};
4     int front = 0, rear = 2;
5     if (front == -1||front > rear)
6         printf("underflow");
7     else
8         printf("delete:%d",q[front++]);
9     return 0;
10 }
```

Test Input

stdin input (optional)

Output

delete:10

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Search



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Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20	Demonstrates complete and precise understanding; identifies all constraints and edge cases	Shows clear understanding with minor gaps	Partial understanding; misses some constraints	Misinterprets problem or lacks clarity
Data Structure Selection	20	Selects optimal data structure with strong justification and comparison	Selects appropriate structure with reasonable justification	Selects somewhat suitable structure with weak reasoning	Incorrect or no data structure selection
Algorithm Design	20	Designs efficient, logically sound, and well-structured algorithm	Algorithm is correct with minor inefficiencies	Algorithm is partially correct or lacks clarity	Algorithm is incorrect or missing
Accuracy of Solution	30	Error-free, well-structured, optimized code/solution	Minor errors but overall functional	Multiple errors; partially working	Non-functional or missing solution
Clarity	10	Exceptionally clear, well-organized, uses diagrams effectively	Clear and organized with minor issues	Somewhat organized but lacks clarity	Poorly organized and unclear

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____